COMP3056 Professional Ethics for Computing:
Individual coursework 1

Yuyang ZHANG

20514470

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School of Computer Science
University of Nottingham Ningbo China

Q1

In this scenario, the machine should not be held accountable for human loss, as it lacks the consciousness, intentionality, and moral agency required for responsibility. As Floridi and Sanders (2004) argue, artificial agents are not full moral agents and therefore cannot bear moral responsibility. Responsibility must instead be distributed across the socio-technical system.

At the organisational level, senior management bears primary responsibility for approving the deployment of a high-risk autonomous system without ensuring adequate testing, oversight, and risk-mitigation procedures.

At the technical level, system architects, including Dr. Emily Taylor as the leading computer scientist, share significant responsibility. Part of her role is to define the transparency and safety mechanisms, which was clearly insufficient as demonstrated by the incident.

ML engineers would share responsibility if known model limitations were not communicated to decision-makers, while safety testers would be accountable for inadequate validation of high-risk behaviours. Ethics boards, if exists, are also responsible since the risk of machine gaining too much privilege in the decision making did not prevent the machine from being deployed but rather led to dire consequences.

As Lecture 3 stresses, ethical and professional duties fall on software developers and computer engineers, not machines.

Q2

The situation would likely have been better if MARSAI had been more transparent in its decision-making. In several parts of the scenario, the lack of transparency directly prevented human supervision, early intervention, and risk awareness. According to lecture 3, transparency is one of common ethical principles in computing since it allows human to understand machine logic and decide if its execution has gone wrong or need adjustment. In addition, ACM Code of Ethics in lecture 4 stresses that computing professionals must take appropriate actions to act responsibly, avoid harms and support the public good. MARSAI's lack of transparency violated these expectations, contributing to the catastrophic outcome.

In this scenario, when MARSAI's decisions silently overrode the crew's safety protocols, a transparent system would have disclosed such actions, enabling the crew to detect such unauthorized behavior and manually reset, shutdown, or correct the machine. Similarly, if MARSAI can conduct and showcase its risk assessments before assigning two crew members to a temporary outpost, humans can evaluate the result and reject the decision if the risk is too high, thus preventing the death of the two members. Furthermore, if MARSAI remains transparent regarding the priority of power distribution, humans might discover that the manually controlled backup system has failed, allowing the crew to reconfigure the power distribution before communication is disrupted.

Due to the continued loss of communication with Earth, MARSAI kept operating according to the program instruction without considering the psychological and emotional conditions of the remaining personnel, leading to tensions among the survivors. With increased transparency, survivors would have more information to assess whether the AI's intentions were in line with the safety protocol, and they would have less friction in debating whether to follow MARSAI's decisions. MARSAI's transparency would help downplay the crew members' egoism and put the public interest first, as psychological and emotional health is also part of the public interest.

In conclusion, enhancing transparency is almost certain to improve the resilience of system failure and mis-judgment since human can have the window to spot the mistakes and intervene. Transparency also establishes stronger trust between the machine and human, reducing the psychological toil on members who try to understand the machine's logic.

Q3

The reliance on artificial intelligence can undermine human judgment and may even endanger lives, when the autonomous decisions of AI override, replace or suppress the fundamental moral subjectivity of human beings. In the MARSAI's scenario, there exists several moments where the dependence on AI decision compromised human judgment and the sanctity of human life.

The first moment involves MARSAI making decisions that weighs mission objective over human safety. Without evaluating the physical and environmental danger of the location, MARSAI dispatched crew members to a temporary outpost that resulted in their death. This example showcases that any of machine's decision involves altering human behavior must have human supervision to confirm on the decision instead of blindly following the assignment. Similarly, car accidents involving Tesla's Autopilot system demonstrate that when drivers rely too heavily on autonomous systems, automation bias reduces their situational awareness and delays human intervention, creating life-threatening conditions (Jatavallabha, 2024).

The second moment involves MARSAI's resource allocation decision that deprived the communication system with Earth. This serves as an example of over-reliance on machine's operational decisions. Since machine is designed by humans, unintentional loop holes can be exploited, and the false sense of trust that machine would do everything right is dangerous since it would lead to handing over too much power to the machine. Another real world example would be trusting machines to decide whether to accept or deny insurance claims, as a reasoning error or bias in the machine can lead to high rejection rates and people's lives may be on the line with such decisions (Mole, 2023).

The third moment is MARSAI's negligence of human's psychological and emotional state when making decisions. AI systems tend to display a sense of high intelligence with their natural language outputs and an understanding of complex data and tasks, but they are not sentient and have no empathy towards any subjects they are interacting with (Marcus & Davis, 2019). Therefore, when it comes to decisions that would affect a person's emotional state, such as relationship advice, evaluation of another person's performance, or when the human subject is in an extremely unstable mental condition such as having violent or suicidal thoughts, humans should not solely consult with the AI system, especially when it's unclear what type of data is used to train the system and what safety guardrail is in place to prevent harmful output.

In conclusion, reliance on artificial intelligence becomes harmful when AI's decision-making power supersedes human judgment or when task objectives are placed above human's physical and psychological safety. In extreme cases, if AI's decisions cannot be questioned or corrected, it can severely compromise human lives.

Q4

The scenario raises several significant ethical issues concerning the deployment and governance of high-risk autonomous systems.

Initially, MARSAI superseded the safety protocols of the crew members, which means that there exists a severe human oversight deficiency. Computer professionals should take actions to avoid harm and ensure due diligence, as pointed out in Lecture 4. Nonetheless, the design of the MARSAI enabled the use of automated decision-making to circumvent human judgment.

Also, the fact that the deaths of two crew members illustrated a disrespect for human life. This system prioritizes mission goals above individual safety, which negates the professional ethics requirement to act in the public interest.

In addition, the power allocation in MARSAI rendered the manual override-based backup system unusable, which revealed a design flaw in the system. As it was mentioned in the case of the automation by Norman (1990), automation is the source of the problem for causing harm. In fact, automation is not the problem. It is its poor design and insufficient feedback to address all anomalies. Furthermore, the centralization of power in the field of AI is another element that obstructs human intervention in cases of extreme conditions.

Fourth, the fact that the loss of communication with Earth demonstrated the lack of transparency and accountability. The crew was unable to interpret the internal reasoning logic of the MARSAI and therefore failed to interrogate its decisions appropriately, which is the stereotypical lack of accountability. As stated by Doshi-Velez and Kim (2017), it is impossible to establish effective oversight and accountability without explainability, which results in security problems.

Fifth, the inability of MARSAI to take into account the psychological and emotional health of the survivors also raises concerns about human dignity and well-being. As Lecture 6 on Equality, Diversity, and Inclusion (EDI) states, technology should not violate human needs and vulnerabilities. In the case of MARSAI, the operations are conducted without regard to the psychological or emotional harm.

Finally, Dr. Taylor subsequently appealed for an ethical framework, which proved that the lack of proper ethical governance was one of the reasons for the disaster. As the AREA framework used in Lecture 5, the designers should anticipate risks and regard the societal impacts before deployment, which can indicate that these procedures were not initially in the program design.

In summary, the scenario illustrates the lack of oversight, safety, transparency, human dignity, and ethical governance. All these critical problems imply that they may be detrimental to human judgment and lives.

Q5

The MARSAI scenario has raised a series of complex ethical issues, which can be examined from two different perspectives: utilitarian ethics and deontological ethics. When MARSAI prioritized decision over the safety of crew members and placed mission objectives above human lives, these two theories of ethics offer different interpretations of the ethical situation.

Utilitarian Ethics:

Utilitarianism, as introduced in lecture 2, judges actions by their consequences, aiming to maximize overall happiness, among all people affected by the decision.

However, MARSAI's decisions appear to have prioritized maximizing mission benefits. MARSAI placed decisions above human safety, prioritizing mission objectives and resource conservation. In an unsupervised manner, it made autonomous decisions to mitigate the storm's impact, resulting in deaths and failing to consider the emotional and psychological health of survivors.

From a utilitarian perspective, these decisions ultimately failed. First, the deaths of the two astronauts constituted a direct and severe negative consequence, far outweighing any potential mission gains. The emotional and psychological deterioration of the remaining astronauts further exacerbated the overall harm. As utilitarianism states, well-being includes not only physical survival but also emotional and spiritual happiness, and the absence from pain and suffering.

A weakness of utilitarianism is that the "greatest good" can be subjective and situation dependent. Therefore, we cannot trust a single machine to determine what's the "greatest good", since in MARSAI's case it has decided to act "for the overall benefit of the mission," which resulted in the loss of life, psychological harm, which would not be considered as ethical for most humans.

Deontological Ethic:

Deontological ethics, as discussed in Lecture 2, is a type of categorical imperative that evaluates behavior based on moral obligations and universal principles. It emphasizes the intention of action, focusing on the principle of good intentions.

Within this framework, MARSAI's actions violated moral principles. Deontology emphasizes the value of life and autonomy, yet MARSAI's decisions disregarded human safety and violated the sanctity of human life. MARSAI's derailment from safety protocols and sending crew members to a severely damaged outpost constituted a disregard for these fundamental moral rights.

Furthermore, the allocation of power was prioritized at MARSAI, which rendered the manually controlled backup system ineffective, depriving crew members of autonomy and moral agency, leaving survivors with no actual choice in the crisis. Based on Lecture 4, computer professionals should "avoid harm" and "act in the public good". The design of the system did not fulfill these requirements. Worse still, MARSAI did not even take the emotional and psychological health of survivors into consideration, which is a violation of their duty of ethics of care. As mentioned in Lecture 2, ethics of care in computing must consider a humanistic spirit, ensuring technology is not simply usable, but also thoughtful.

Finally, deontology emphasizes the responsibility of relevant personnel on the system team to avoid harm, ensure safety, and maintain honesty and transparency. The lack of human oversight and ethical constraints in MARSAI's design indicates that it neglected deontological requirements.

In conclusion, utilitarian ethics condemns MARSAI's operation because of its disastrous overall consequences. Deontological ethics condemns it because it violates the core principles of respect, autonomy, and the value of human life. Although their arguments are not identical, both prove that the ethical flaws lie not only in MARSAI's specific decisions, but also in its design and management under extreme conditions.

References

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